YUHANG JIANG

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Summary

- EIT Manufacturing double-degree master's in Computer Science (UNITN) and Data Science (SUPSI).
- First author of five 2026 papers (three sole-authored) across embodied AI, mechanistic interpretability of state-space models, adversarial robustness, and remote sensing.
- Research intern at Huawei Pisa Research Center and FBK.
- Kaggle Competitions Expert.

Education

EIT Manufacturing Master & Doctoral School (Double Degree) Sep 2023 - Mar 2026
Data Science and AI for a Competitive Manufacturing
- **Università di Trento (UNITN), Italy**
  MSc, Computer Science · GPA 98/110
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI), Switzerland**
  Master of Engineering (MSE), Data Science · GPA 5.1/6.0
- **Scholarship:** Recipient of the EIT Manufacturing Scholarship (10,000 EUR/yr)

Anhui University (Project 211), China Sep 2019 - Jun 2023
BEng, Intelligent Science and Technology
- **Average Score:** 84.34/100
- **Awards:** Second Prize for Outstanding Academic Performance (2022); First Prize Scholarship of Academic Science and Technology (2021)
- **Courses:** Machine Learning, Pattern Recognition, Big Data Analysis, Numerical Analysis, Natural Language Processing.

Publications

"PInVerify: An Offline Embodied Benchmark for Active Instance Verification"
Poster, FMEA Workshop @ CVPR 2026, Project Page, arXiv:2605.30639, OpenReview
Master's thesis project.
Proposes Active Instance Verification, a new embodied task. An agent actively selects viewpoints to confirm a fine-grained object description before committing to a decision. On-device-scale MLLM agents fine-tuned via LoRA and RL.
Yuhang Jiang (sole author)

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"
IEEE TGRS, under review, Project Page, arXiv:2607.03869
A training-free neuro-symbolic method that synthesizes and executes typed spatial programs over vision-language models to resolve referring expressions in remote-sensing images. It more than doubles the best prior training-free method on RRSIS-D.
Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"Task Structure Reverses Layerwise State Encoding in Sequence Models"
Under review, arXiv:2606.00926
Shows that layerwise state encoding in sequence models reverses with task structure, via probing and causal intervention across Transformers, Mamba, and RNNs.
Yuhang Jiang (sole author)

"Detection vs. Execution: Single-Bucket Probes Miss Half the Mamba-2 State Sink"
Under review, arXiv:2606.00930
Shows single-bucket probes miss much of the Mamba-2 state sink's causal substrate: representational similarity does not imply functional equivalence.
Yuhang Jiang (sole author)

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"
Under review, arXiv:2606.22516
Resize-based input diversity helps transfer attacks from standard surrogates but hurts robust ones, explained through gradient geometry.

Yuhang Jiang, Xiaojing Chen

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"
WRC SARA 2025 (oral)
Bowen Zhang, **Yuhang Jiang**, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"
CAIBDA 2022 (oral)
Yuhang Jiang*, Wenping Tong

Software Copyright of Intelligent contract software for agricultural insurance compensation
2022SR1568111.
*Note: * indicates the corresponding author.*

Academic Service

Reviewer, FMEA Workshop @ CVPR 2026.

Work Experience

Huawei Pisa Research Center Apr 2026 - Oct 2026
Research Intern Pisa, Italy
- Working on memory systems for LLM agents, focused on long-horizon, context-persistent behaviour.
- Conducting LLM post-training, applied to cockpit AI.

Fondazione Bruno Kessler - FBK Mar 2025 - Mar 2026
Research Intern Trentino-Alto Adige, Italy
- Proposed a new embodied perception task and built the first offline multi-view benchmark for active instance verification in realistic 3D environments.
- Designed an MLLM-based agent with attribute decomposition, confidence-weighted multi-view tracking, and next-best-view planning; fine-tuned via LoRA with SFT followed by RL alignment, achieving 85.6% accuracy.

Chengdu Jiaoda Guangmang Technology Co., Ltd. Jul 2022 - Sep 2022
Algorithm Intern Chengdu, Sichuan, China
- Researched the latest machine learning and industrial anomaly detection algorithms.
- Applied GAN models to reconstruct images to identify anomalies and evaluate the performance.
- Used the object detection neural network model to detect abnormal parts in high-speed rail supports.
- Assisted colleagues in developing a set of character recognition patents for LCD meters.

Selected Projects

KDEN-NSGA-II Scheduler for Flexible Job-Shop Scheduling Jun 2025

- Developed a multi-objective FJSP solver based on enhanced NSGA-II with hybrid initialization and adaptive crossover, optimizing makespan, load balance, and machine idle time.
- Built a modular benchmarking framework supporting multiple baselines (NSGA-II, NSGA-III, SPEA2, multi-objective ACO) with KNN-based density selection for diversity-aware search.

High Performance Computing for Grey Wolf Optimizer (GWO) Optimization Jan 2025

- Proposed HGT-GWO, integrating global historical best positions and trend guidance into GWO, outperforming standard GWO on 15 benchmark functions.
- Implemented a master-worker island parallelization scheme in C/MPI/OpenMP on UNITN's HPC cluster, reducing communication overhead through controlled synchronization.

Augmented Reality-Driven Robotic Arm Control for Industrial Automation Sep 2024

- Built an AR-based robotic arm control system on Microsoft HoloLens 2, combining YOLOv8 and FastSAM for real-time gesture recognition and fingertip coordinate mapping.
- Supported multi-mode operation (gesture-based and interface-based) with robustness to hand occlusion in industrial environments.

Research on Semantic Segmentation Method of High-Resolution Remote Sensing Images May 2023
Outstanding Undergraduate Graduation Project
- Developed an encoder-decoder model with residual-weighted attention for remote sensing segmentation, achieving F1-scores of 90.23% and 87.37% on ISPRS Potsdam and Vaihingen datasets.

Competitions

Kaggle March Machine Learning Mania 2025 - Silver Medal (64th / 1,727) Apr 2025
Solo

Kaggle BirdCLEF 2024 Competition - Bronze Medal (70th / 974) Jun 2024
Solo

Kaggle BirdCLEF+ 2025 Competition - Bronze Medal (199th / 2,025) Jun 2025
Solo

Mathematical Contest In Modeling (MCM), Finalist winner (top 1%) Apr 2021
Leader

- Designed a set of algorithms to assess the degree of hunger around the world and optimize the food supply chain.
- Responsible for modeling, programming and part of paper writing.
- The youngest winner in the history of Anhui University and the first winner of School of Internet.

iCAN Innovation Contest 2021 (Anhui) - Second prize Oct 2021
Team member

- Developed a set of hardware system capable of intelligently monitoring and warning the abnormal behavior of the elderly.
- Responsible for some STM32 development and project plan writing.

Skills

- **Programming:** Python, C/C++, Java, Matlab, R, C#.
- **ML / DL Frameworks:** PyTorch, scikit-learn, TensorFlow, PaddlePaddle; MPI / OpenMP (HPC).
- **LLM / VLM:** multimodal LLMs (MLLMs), post-training (SFT, DPO, RL alignment incl. GRPO/GSPO), LoRA / PEFT, RAG, LLM agents.
- **Architectures:** Transformer, Mamba, CNN, RNN.
- **Areas:** Computer Vision, NLP, Reinforcement Learning, Mechanistic Interpretability, Evolutionary Algorithms.
- **Applications:** Image Classification & Segmentation, Object Tracking, Industrial Anomaly Detection, Time-Series Analysis.
- **Other:** Unity; Lean Manufacturing & Sustainable Management.

Language

- **English:** Fluent, TOEFL 102
- **Chinese:** Native Speaker